

memory required is dependent on the complexity of the design, where it shares only the recommended amount.

Competitors at hand

Ashley Newson, the creator of SmartSim says, "SmartSim is not a fully-fledged printed circuit board (PCB) or other hardware design tool like KiCad, quite universal circuit simulator (Qucs) or SPICE, and so cannot be compared to them." As per Newson, the only software we can compare SmartSim to is Logisim.

Logisim is another educational tool for designing and simulating digital logic circuits. It is again an open source software, with colour-coded wiring that aids during simulation. It runs on all devices that support Java version 5 or later, and has a special version released for MacOS X and Windows OS.

Although it shares a high resemblance with SmartSim, Logisim cannot accommodate any analogue components, and any further development to it has also been suspended indefinitely.

Another difference is SmartSim's compatibility with Raspberry Pi, which has not yet been proved in case of Logisim.

User experiences

We are all humans and, despite being aware of the features and advantages of this software, we want to know the experiences of other users too.

As per user reviews on SmartSim blog created by Raspberry Pi, one of the reviewer uses it on three different platforms and was highly impressed by its multi-platform capability. As per him, SmartSim looks and feels identical on all three of them and creates circuit

Quick view

- Version: 1.4.1 (Latest)
- Operating systems and processor architectures (tested)
 - Windows XP or later (x86)
 - Linux based (x86; x86_64)
 - ARM
- Licence: GNU general-public licence Version 3
- Free disk space: 25MB (including libraries)
- Free memory: 25MB
- Peripherals
 - Mouse, touch-screen or other pointing device
 - Keyboard
 - Monitor or other display
 - Optional printer

Colour description

- Blue wire: Set to binary 0 (false)
- Red wire: Set to binary 1 (true)
- Green wire: Does not have a live input (high impedance, Z or state). This can also be used as a result of a fault design with no input connections to a wire
- Yellow wire: Multiple-input error

SmartSim on Raspberry Pi

Before using SmartSim on Raspberry Pi, it has to be upgraded to the latest version and then installed by typing the following commands:

- `sudo apt-get update`
- `sudo apt-get install smartsim`

Pre-requirements

SmartSim requires a few tools without which it cannot be installed on different platforms

- GTK+ 2
- LibRsvg

files that can be transferred between different versions seamlessly.

Another user, too, accepted that the software is pretty good. He further added that, SmartSim is slick, easy-to-use, looks great and seems to do exactly what it says on the package. He said that, he was surprised by the performance of SmartSim on the fairly-large circuit he connected.

Although this tool has had almost all positive feedback, one customer was disappointed because he could not install it on his old Fedora system. Newson agreed that there was a small bug in the .make file of the software that he removed and then the installation went perfectly. ●

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